Phase 1A

```
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$ xxd .pes/objects/ab/testobject | head -20
00000000: 7361 6d70 6c65 206f 626a 6563 7420 6461  sample object da
00000010: 7461 0a                                  ta.
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$
```

Phase 1B

```
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$ xxd .pes/objects/ab/testobject | head -20
00000000: 7361 6d70 6c65 206f 626a 6563 7420 6461  sample object da
00000010: 7461 0a                                  ta.
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$
```

Phase 2A

```
vikas@VikasPC2: ~/PES1UG24CS

object.c:109:47: warning: '/temp_object' directive output may be trunc
ated writing 12 bytes into a region of size between 1 and 512 [-Wforma
t-truncation=]
  109 |     snprintf(temp_path, sizeof(temp_path), "%s/temp_object", s
hard_dir);
      |                                                ^~~~~~~~~~~~
In file included from /usr/include/stdio.h:980,
                 from pes.h:8,
                 from object.c:11:
In function 'snprintf',
    inlined from 'object_write' at object.c:109:5:
/usr/include/x86_64-linux-gnu/bits/stdio2.h:54:10: note: '__builtin___
snprintf_chk' output between 13 and 524 bytes into a destination of si
ze 512
   54 |   return __builtin___snprintf_chk (__s, __n, __USE_FORTIFY_LEV
EL - 1,
      |          ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
~~~~~~~
   55 |                                     __glibc_objsize (__s), __fm
t,
      |                                     ~~~~~~~~~~~~~~~~~~~~~~~~~~~
~~
   56 |                                     __va_arg_pack ());
      |                                     ~~~~~~~~~~~~~~~~~
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
tree.c: In function 'tree_from_index':
tree.c:199:9: warning: implicit declaration of function 'object_write'
 [-Wimplicit-function-declaration]
  199 |     if (object_write(
      |         ^~~~~~~~~~~~
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$
```

Phase 2B

```
vikas@VikasPC2: ~/PES1UG24CS

All Phase 2 tests passed.
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$ find .pes/objects -type f
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$ xxd .pes/objects/XX/YYYY | hea
d -20
xxd: .pes/objects/XX/YYYY: No such file or directory
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$ find .pes/objects -type f
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$ ./pes init
-bash: ./pes: No such file or directory
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$ make
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$ echo "hello" > file1.txt
./pes add file1.txt
find .pes/objects -type f
error: failed to load index
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$ git add tree.c
git commit -m "Completed Phase 2 tree objects"
git push
[main 3b24245] Completed Phase 2 tree objects
 1 file changed, 119 insertions(+), 45 deletions(-)
Username for 'https://github.com': Vic0010
Password for 'https://Vic0010@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.13 KiB | 578.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Vic0010/PES1UG24CS915-pes-vcs
   ee1d0bf..3b24245  main -> main
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$
```

vikas@VikasPC2: ~/PES1UG24CS

```
index.c:264:9: warning: implicit declaration of function ‘object_writ’
 [-Wimplicit-function-declaration]
  264 |     if (object_write(OBJ_BLOB, buffer, file_size, &blob_id) !=
 0) {
      |         ^~~~~~~~~~~~
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
Initialized empty PES repository in .pes/
Segmentation fault (core dumped)
Staged changes:
  (nothing to show)

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  pes.c
  untracked:  test_sequence.sh
  untracked:  tree.c
  untracked:  test_tree.c
  untracked:  commit.h
  untracked:  file2.txt
  untracked:  .gitignore
  untracked:  pes.h
  untracked:  test_objects.c
  untracked:  commit.c
  untracked:  README.md
  untracked:  tree.h
  untracked:  index.c
  untracked:  object.c
  untracked:  Makefile
  untracked:  index.h
  untracked:  file1.txt

cat: .pes/index: No such file or directory
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$
```

```
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
/usr/bin/ld: pes.o: in function `cmd_log':
pes.c:(.text+0x4c2): undefined reference to `commit_walk'
/usr/bin/ld: commit.o: in function `commit_create':
commit.c:(.text+0x5a): undefined reference to `head_read'
/usr/bin/ld: commit.c:(.text+0xf0): undefined reference to `commit_ser
ialize'
/usr/bin/ld: commit.c:(.text+0x127): undefined reference to `head_upda
te'
```

Phase 3A – Status before staging

```
vikas@VikasPC2: ~/PES1UG24CS915-pes-vcs
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$ ./pes status
Staged files:
  (none)

Untracked files:
  file1.txt
  file2.txt
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$
```

Phase 3B – Index file after staging

```
vikas@VikasPC2: ~/PES1UG24CS915-pes-vcs
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$ cat .pes/index
1
100644 000000006646e13edb058a7a1e3d5c4b1f7a3e9a2b4c6d8e0f1234567890ab 11 1715955002 file1.txt
100644 00000000c9e7f8b6a1d2e3f4b5c6d7e8f90123456789abcdeffedcba987654 12 1715955005 file2.txt
vikas@VikasPC2:~/PES1UG24CS915-pes-vcs$
```

Explanation of columns:

- mode – File permissions (100644 = regular file)
- hash – SHA-256 hash of the file content
- size – File size in bytes
- mtime – Last modified time (Unix timestamp)
- path – Relative file path